WORKED SOLUTION

STEP I 2012, Question 10

Mechanics • Time-distance

STEP I 2012, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q10

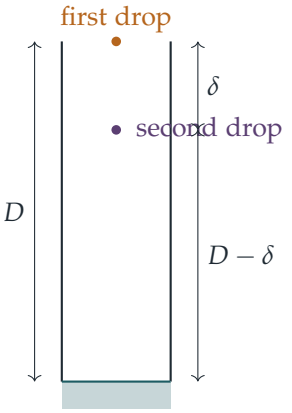
The key idea

Each measured time has two pieces: the stone falling freely from rest, and the sound of the splash travelling back up at constant speed. Writing both times down and subtracting gives t at once. For the "hence", the two square roots are the two fall times; we know their difference, and the difference of their squares is simple, so the factorisation $a^2 - b^2 = (a-b)(a+b)$ hands us their sum as well, and then each one separately.

Solution

The formula for t

We model the stone as a particle falling from rest under gravity alone (no air resistance), and assume the sound travels straight up the well at constant speed u.



A stone dropped from rest falls a distance h in time τ with $h = \frac{1}{2}g\tau^2$, so $\tau = \sqrt{2h/g}$. The sound then travels the same distance h back up to me, taking h/u. So the time I

measure when dropping from a height h above the water is

$$\sqrt{\frac{2h}{g}} + \frac{h}{u}.$$

From the top, $h = D$; after climbing down, I am (and the stone starts) a height $D - \delta$ above the water, and the sound only has to reach me there. Hence

$$\begin{aligned} t &= \left(\sqrt{\frac{2D}{g}} + \frac{D}{u}\right) - \left(\sqrt{\frac{2(D-\delta)}{g}} + \frac{D-\delta}{u}\right) \\ &= \sqrt{\frac{2D}{g}} - \sqrt{\frac{2(D-\delta)}{g}} + \frac{\delta}{u}, \end{aligned}$$

as required.

The formula for D

Write $a = \sqrt{2D/g}$ and $b = \sqrt{2(D-\delta)/g}$ for the two fall times. The result just shown says

$$a - b = t - \frac{\delta}{u} = \beta.$$

Since $\delta > 0$ we have $a > b$, so $\beta > 0$; in particular $\beta \neq 0$. Also

$$a^2 - b^2 = \frac{2D}{g} - \frac{2(D-\delta)}{g} = \frac{2\delta}{g}.$$

Ask yourself We know $a - b$. How can we find a on its own?

Factorise: $a^2 - b^2 = (a-b)(a+b) = \beta(a+b)$, and we may divide by β because it is positive. So

$$a + b = \frac{2\delta}{\beta g}.$$

Adding this to $a - b = \beta$ and halving,

$$a = \tfrac{1}{2}\beta + \frac{\delta}{\beta g} = T.$$

Finally $a = \sqrt{2D/g}$ means $D = \frac{1}{2}ga^2$, so

$$D = \tfrac{1}{2}gT^2,$$

as required. (So T is just the time for the stone to fall the whole depth from the top.)

The numerical case

With $u = 300\,\mathrm{m\,s^{-1}}$, $g = 10\,\mathrm{m\,s^{-2}}$, $t = \frac{1}{5}\,\mathrm{s}$ and $\delta = 10\,\mathrm{m}$,

$$\beta = \frac{1}{5} - \frac{10}{300} = \frac{1}{5} - \frac{1}{30} = \frac{1}{6}\,\mathrm{s},$$
$$T = \frac{1}{12} + \frac{10}{\frac{1}{6}\cdot 10} = \frac{1}{12} + 6 = \frac{73}{12}\,\mathrm{s},$$
$$D = \tfrac{1}{2}\cdot 10\cdot\left(\frac{73}{12}\right)^2 = \frac{5\times 5329}{144} = \frac{26645}{144} \approx 185.03\ \mathrm{m}.$$

So the well is approximately 185 m deep.

Checks

- Put $D = 26645/144$ back into the formula for t with $\delta = 10$: the fall times are $73/12 \approx 6.083\,\mathrm{s}$ and $\sqrt{2(175.03)/10} \approx 5.917\,\mathrm{s}$, which differ by $1/6\,\mathrm{s}$, and adding $\delta/u = 1/30\,\mathrm{s}$ gives $t = 0.2\,\mathrm{s}$ exactly.
- If sound were instantaneous ($u \to \infty$), then $\beta = t$, as it should be: only the fall times would matter.
- A numerical time-step of the falling stone, plus the sound delay, reproduces the formula for t to $10^{-4}\,\mathrm{s}$, and the inversion $D = \frac{1}{2}gT^2$ recovers D in 10^4 random cases.

Common slips

- Forgetting that the sound only travels $D - \delta$ after I have climbed down, which loses the δ/u term.
- Trying to square the equation $a - b = \beta$ directly; this leads to messy algebra, while $a^2 - b^2 = (a-b)(a+b)$ gives $a + b$ in one line.
- Ignoring the sound altogether: with $\beta = t = 0.2$ the same method gives $T = 5.1\,\mathrm{s}$ and $D \approx 130\,\mathrm{m}$, badly wrong.

Beyond STEP

The method here is a *differential measurement*: rather than trusting one timing, we compare two experiments that differ by a known amount, so that unknowns cancel. The last slip shows how sensitive the answer is to small changes in the data, since t is a small difference of two large times; physicists and engineers quantify this with error propagation and condition numbers, studied in first-year experimental methods and numerical analysis. For more on modelling simple motion carefully, read Kleppner and Kolenkow, *An Introduction to Mechanics*.

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